

Osaid Khan

[+1 \(945\) 304-5781](tel:+19453045781) | khanosaid726@gmail.com | www.linkedin.com/in/-osaid-khan/ | github.com/ozzyozbourne

Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the **Software Engineer II** role with **Didi**. I bring over three years of professional experience in developing highly-scalable backend systems and simulation frameworks, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY.

Through my experience across engineering roles, I have built production-grade systems that directly reflect **the development of highly-scalable robotic simulation systems and autonomous driving algorithms**. As a Software Engineer at Sperse, I architected a highly-scalable orchestration layer using **Python** and **C++** for 1000+ endpoints — requiring the same level of architectural rigor and performance optimization needed to run complex simulation jobs in the cloud. To ensure system reliability, I implemented a robust observability and metrics layer using Langfuse and LangSmith, which enabled real-time software performance analysis and tracing. This focus on data-driven metrics directly mirrors the responsibilities at Didi for designing scenarios and metrics to predict real-world algorithm performance. Previously, as a Software Engineer at Qualitest, I designed a highly-scalable cloud-agnostic storage service to handle high-concurrency simulation data across AWS S3 and GCS, and utilized **Apache Kafka** and **Apache Spark** for real-time event streaming. I also developed backend workflows using AWS Step Functions and **Docker** to handle long-running simulation jobs, ensuring the kind of operational stability and system-level thinking **this role demands**. Together, these experiences have prepared me to **contribute to the design of scalable software for cloud-based simulation and performance analysis** and drive impact within Didi's autonomous driving team.

My academic research experience has provided a rigorous foundation in generative modeling and technical communication — skills that translate directly into **the data-driven innovation and “Safety First” engineering discipline Didi values**. As an AI Researcher at the Pace Artificial Intelligence Lab, I conducted research on **robotic simulation systems** using StyleGAN and Stable Diffusion to generate high-fidelity synthetic data for performance verification. I performed deep software performance analysis and optimization of these models on **Linux OS** using **C++** and CUDA, ensuring efficient and highly-scalable inference across distributed nodes. Beyond technical implementation, I facilitated workshops on Linux environments and autonomous agent design for an audience of 50+ students. Breaking down complex technical concepts for diverse audiences mirrors **the collaborative scenario design and technical communication responsibilities central to this role**. This experience has shaped me into an engineer who builds with rigor and communicates with clarity, both of which I would bring to **Didi and its mission**.

I have always been passionate about **“Building a better journey together”** and the prioritization of **“Safety First”** in autonomous mobility and envision myself contributing my talent to the furthering of **transforming the future of shared mobility through intelligent and sustainable transportation**. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration, I look forward to connecting!

Sincerely,
Osaid Khan